

Ipsita Ray

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Research interests

The field of my research is Particle Physics Phenomenology. I have worked in the field of heavy flavor physics, specifically the data-driven analysis of new physics effects in the semi-leptonic B-meson decays via the neutral and charged current exchanges in a model-independent effective field theory (EFT) approach, the precision extraction of the Cabibbo-Kobayashi-Maskawa (CKM) elements and studies of physics beyond the Standard Model of Particle Physics. I have also worked on the form factor calculations within the Light Cone Sum Rule (LCSR) approach. I have strong interests and experience in the computational aspects like statistical inference, machine learning, data science, and programming in Python, Mathematica and C++. Some of my recent research interests include:

- Application of flavor symmetries to the study of non-leptonic decays using Group Theory methods
- Dark matter and neutrino physics
- Application of Quantum Information Science in the domain of High Energy Physics
- Application of the machine learning and AI techniques to various aspects of physics

Academic appointments

2024- June 2026	Postdoctoral fellow :	Universite de Montreal
	Postdoc supervisor :	Prof. David London
2022-2024	Postdoctoral fellow :	Indian Institute of Technology Gandhinagar
	Postdoc supervisor :	Prof. Rusa Mandal

Education

2017-2022	Ph.D. in Physics :	Indian Institute of Technology Guwahati
	Ph.D. supervisor :	Prof. Soumitra Nandi
	Thesis title :	Some studies of the semileptonic B -meson decays via the neutral and charged current exchanges.
	Year of award :	July, 2023
2014-2016	M.Sc in Physics :	University of Calcutta, India
2011-2014	B.Sc in Physics :	University of Calcutta, India

Research experience -

Published or arXived manuscripts

- L.Kolay, S.Nandi, I.Ray, **Benchmark Dark SMEFT Scenarios from the Interplay of Flavor Physics and Dark Matter Searches**, arXiv:2609.26886
- B.Bhattacharya, M.Bouchard, A.Jean, D.London, I.Ray, **Anomalies in Hadronic $B \rightarrow VV$ Decays**, arXiv:2607.27202
- B.Bhattacharya, M.Bouchard, A.Jean, D.London, I.Ray, **Isospin-based EWP-tree Relations**, JHEP 05 (2026) 195
- MSAA.Khan, R.Mandal, P. Patil, I.Ray, **Investigating non-local contributions in $B_s \rightarrow \phi \bar{l}l$ including higher-twist effects**, JHEP 07 (2025) 068
- R.Mandal, P. Patil, I.Ray, **Probing the inverse moment of B_s -meson distribution amplitude via $B_s \rightarrow \eta_s$ form factors**, JHEP 06 (2024) 212.
- R.Mandal, S.Nandi, I.Ray, **Constraining inverse moment of B -meson distribution amplitude using Lattice QCD data**, Phys.Lett.B 848 (2024) 138345.
- I.Ray and S.Nandi, **Test of new physics effects in $\bar{B} \rightarrow (D^{(*)}, \pi)\ell^- \bar{\nu}_\ell$ decays with heavy and light leptons**, JHEP 01 (2024) 022.
- A.Biswas, S.Nandi and I.Ray, **Extractions of $|V_{ub}|/|V_{cb}|$ from a combined study of the exclusive $b \rightarrow u(c)\ell\nu_\ell$ decays**, JHEP 07 (2023) 024.
- A.Biswas, S.Nandi, S.K.Patra and I.Ray, **Study of the $b \rightarrow d\ell\ell$ transitions in the Standard Model and test of New Physics sensitivities**, JHEP 03 (2023) 247.
- A.Biswas, S.Nandi, S.K.Patra and I.Ray, **A closer look at the extraction of $|V_{ub}|$ from $B \rightarrow \pi l\nu$** , JHEP 07 (2021) 082.
- A.Biswas, S.Nandi, S.K.Patra and I.Ray, **New physics in $b \rightarrow sll$ decays with complex Wilson coefficients**, Nucl.Phys.B 969 (2021) 115479.

Conference proceedings

- I.Ray, **New Physics effects in $b \rightarrow sll$ decays**, Springer Proc.Phys. 277 (2022) 127-131.
- I.Ray, **A relook into the extraction of exclusive $|V_{ub}|$** , PoS ICHEP2022 714
- I.Ray, **Extraction of $|V_{ub}|$ from $B \rightarrow \pi l\nu$ modes**, Frascati Phys.Ser. 72 (2022) 80-85

Review

- Reviewer of European Physical Journal C

Oral Presentations

- **Anomalies in Hadronic B decays**
33rd International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2026), Norman, Oklahoma, 2026
- **Analysis of $B \rightarrow VV$ decays under $SU(3)_F \otimes SU(2)_{\text{spin}}$**
32nd International Symposium on Lepton Photon Interactions at High Energies (Lepton Photon 2025), Madison, Wisconsin, 2025
- **Analysis of $B \rightarrow VV$ decays under $SU(3)_F \otimes SU(2)_{\text{spin}}$**
23rd Conference on Flavor Physics and CP violation (FPCP 2025), Cincinnati, Ohio, 2025
- **Constraining inverse moment of B -meson distribution amplitude using Lattice QCD data**
4th Heavy Flavor Meet, IIT Goa, India, 2023

- **Semileptonic B -meson decays via the neutral and charged current exchanges**
Workshop on Heavy Hadrons in heavy ion and particle collisions (HHHPCS 2023), IIT Gandhinagar, India, 2023
- **Is the Standard Model a complete theory?**
Bangabashi Evening College, Kolkata, India, 2022
- **Extraction of $|V_{ub}|$ from $B \rightarrow \pi l \nu$**
XX LNF Summer School in Nuclear, Subnuclear and Astroparticle Physics, LNF, Italy, 2022
- **A relook into the extraction of exclusive $|V_{ub}|$**
41st International Conference on High Energy Physics (ICHEP 2022), Bologna, Italy, 2022
- **New physics in neutral current $b \rightarrow s$ decays**
The University of Warwick, 2022 (online mode)
- **New physics in $b \rightarrow sll$ decays with complex Wilson coefficients**
30th International Symposium on Lepton Photon Interactions at High Energies, 2021 (online mode)
- **New physics in $b \rightarrow sll$ decays with complex Wilson coefficients**
Anomalies, 2021 on Flavour anomalies (online mode)
- **A closer look at the extraction of $|V_{ub}|$ from $B \rightarrow \pi l \nu$.**
The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021) (online mode)
- **New Physics effects in $b \rightarrow sll$ decays.**
XXIV DAE-BRNS Symposium on High Energy Physics, 2020 (online mode)

Invited talks

- **Extraction of the ratio $|V_{ub}|/|V_{cb}|$ from a combined study of the exclusive decays**
Invited talk at the 12th International Workshop on the CKM Unitarity Triangle (CKM 2023), Santiago de Compostela, Spain, 2023
- **Analysis of the rare $b \rightarrow dll$ transitions within and beyond the Standard Model**
Invited seminar at Karlsruhe Institute of Technology, 2023 (online mode)

Conferences, Workshops and Schools attended

- **AI Reasoning in Theoretical Physics**, Aspen, Colorado, May 24-June 07, 2026
- **33rd International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2026)**, Norman, Oklahoma, May 18-22, 2026
- **32nd International Symposium on Lepton Photon Interactions at High Energies**, Madison, Wisconsin, August 25-29, 2025
- **23rd Conference on Flavor Physics and CP violation (FPCP 2025)**, Cincinnati, Ohio, June 2-6, 2025
- **School for Advanced Topics in Particle Physics (SATPP)**, ICTS, India, April 8-19, 2024
- **XVII Workshop on High Energy Physics Phenomenology (WHEPP 2024)**, IIT Gandhinagar, India, January 2-11, 2024
- **4th Heavy Flavor Meet**, IIT Goa, India, November 1-4, 2023
- **12th International Workshop on the CKM Unitarity Triangle (CKM 2023)**, Santiago de Compostela, Spain, September 18-22, 2023
- **Workshop on Heavy Hadrons in heavy ion and particle collisions (HHHPCS 2023)**, IIT Gandhinagar, India, March 24-25, 2023
- **Applied Statistical Modeling and Data Analytics for Petroleum Engineering and Related Applications**, IIT Guwahati, India, November 22-24, 2022

- **XX LNF Summer School in Nuclear, Subnuclear and Astroparticle Physics, 2022**, LNF, Italy, July 11-15, 2022
- **41st International Conference on High Energy Physics (ICHEP 2022)**, Bologna, Italy, July 6-13, 2022
- **30th International Symposium on Lepton Photon Interactions at High Energies**, January 10-14, 2022
- **Anomalies, 2021 on Flavour anomalies**, IIT Hyderabad, India, November 10-12, 2021
- **The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021)**, August 23-28, 2021
- **XXIV DAE-BRNS Symposium on High Energy Physics, 2020**, December 14-18, 2020
- **XXXIII Science and Engineering Research Board (SERB) Main School in Theoretical High Energy Physics, 2019**, University of Delhi, India, December 7-26, 2019
- **XVI Workshop on High Energy Physics Phenomenology (WHEPP 2019)**, IIT Guwahati, India, December 1-10, 2019
- **Madgraph School on the collider related package (Madgraph)**, IMSc Chennai, India, November 18-22, 2019
- **Anomalies, 2019 on Flavour anomalies**, IIT Hyderabad, India, July 18-20, 2019

Teaching experience -

- **PH-210, 2021**, Teaching assistant for laboratory courses to the 2nd year MSc students, IIT Guwahati
- **PH-512, 2020**, Teaching assistant for the course on “Classical mechanics” to 1st year MSc students, IIT Guwahati
- **PH-503, 2020**, Teaching assistant for the course on “Atoms and molecules” to 1st year MSc students, IIT Guwahati
- **PH-110, 2019**, Teaching assistant for laboratory courses to the 1st year BTech students, IIT Guwahati
- **PH-110, 2018**, Teaching assistant for laboratory courses to the 1st year BTech students, IIT Guwahati

Mentorship experience -

- **Praveen S Patil** (IIT Gandhinagar, 2023-2024): mentoring in two publications.

Computational Skill -

- Programming language: Python, C++
- Software: Mathematica, Latex
- Frequentist and Bayesian Statistical Analysis
- Theoretical knowledge: Quantum field theory, Quantum mechanics, Group theory, Effective field theory
- Packages
 - OptEx (mathematica package for Statistical Analysis)
 - Minuit (Python interface for Statistical Analysis)
 - Dynesty (Python package for Bayesian inference)
 - MadGraph, Pythia and MadAnalysis (collider related package)
 - FeynCalc (mathematica package for form factor calculations and scattering amplitudes)
 - FeynRules (mathematica package for writing specifics of various models)
 - Wilson (python package for running Wilson coefficients from one scale to other)
 - Package-X (mathematica package for loop calculation)
 - Pandas, numpy, sklearn, tensorflow (data analytics and machine learning)

Honors and Awards

- Simons Foundation International grant to attend the workshop titled “AI Reasoning in Theoretical Physics” in Aspen, Colorado.
- SERB-International Travel Support (ITS) grant by the Government of India to attend the CKM 2023 workshop at Santiago de Compostela, Spain
- Student Travel Allowance Fund (STAF) from Indian Institute of Technology Guwahati to attend ICHEP 2022 conference at Bologna, Italy
- National Eligibility Test (NET LS) 2018
- Graduate Aptitude Test in Engineering (GATE) 2016 - All India Rank 244
- Joint Entrance Screening Test (JEST) 2016 - All India Rank 402

Professional services and activities

- Member of the organizing committee for **XVII Workshop on High Energy Physics Phenomenology (WHEPP 2024), IIT Gandhinagar, India**
- Member of the organizing committee for **Workshop on Heavy Hadrons in heavy ion and particle collisions (HHHPCS 2023), IIT Gandhinagar, India**
- Member of the organizing committee for **XVI Workshop on High Energy Physics Phenomenology (WHEPP 2019), IIT Guwahati, India**

References

Prof. David London

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